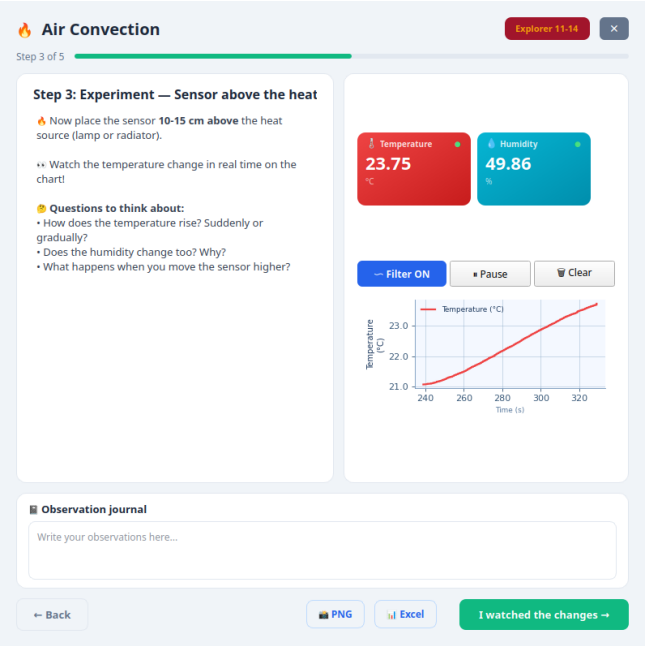
# Air Convection

Why warm air rises, and how to see an invisible current with a temperature sensor

**Level:** Explorer — ages 11–14 (Romanian school years V–VIII) • **Time in class:** 20 min, two 50-minute lessons or three 50-minute lessons • **Sensors:** BME280 (temperature, humidity, pressure)




*The activity screen. The values in the picture are simulated; your own will differ.*

## Before you start

| | |
|---|---|
| **Who it is for** | Explorer — ages 11–14 (Romanian school years V–VIII). The short route also works in year IV as a demonstration, without the density calculation. The most direct syllabus entry is **year VIII**, where heat transfer by conduction, convection and radiation is explicit material. |
| **Assumed knowledge** | Reading a graph over time, calculating a difference, and dividing to get a rate (degrees per metre). For the extension, in addition: density as mass per volume and working with percentages. For support: only “warmer / cooler” and “up / down”. |
| **Time in class** | Short: 20 min. Full: two 50-minute lessons (1 h 40 min). Project: three 50-minute lessons (2 h 30 min). |
| **Teacher preparation** | 15 min — **testing the heat source**, taping the wall at the five heights, cutting the paper strips, closing the windows half an hour beforehand, printing the sheets. |
| **Sensor settling time** | **60–120 seconds** for temperature after every move; humidity is slower, often three minutes. It is the single most common reason for bad data. |
| **Sensors and boxes** | BME280. One box per group; at least two groups, so you can compare two heat sources — ideally a wet one (the container of warm water) and a dry one (the radiator, a filament lamp or the laptop’s air vent). |
| **Three routes** | |

| | |
|---|---|
| **Short** | 20 min: exactly the app's wizard — the prediction with arrows, the reference at desk height, above the source, the second prediction, the return, how sure we are, the conclusion. |
| **Full** | Two 50-minute lessons: lesson 1 is the wizard, unhurried, with the loop drawing; lesson 2 is the vertical profile at five heights and the class graph. |
| **Project** | Three 50-minute lessons: the two above, plus the applications lesson — density, the sea breeze, the hot-air balloon, the written explanation. |

## 1. Where this fits in the curriculum

Convection is one of the few topics students have heard about since primary school but for which they have never seen a number. Here they measure it themselves, in their own classroom, where nothing special is happening. The activity fits:

- **Physics** (*Fizică*) — at **year VIII**, where thermal phenomena cover heat and **heat transfer by conduction, convection and radiation**, the activity sits exactly on top of the syllabus: convection is distinguished from the other two by the fact that *the substance itself moves*. At **year VI** it connects to temperature, thermal equilibrium and the quantities mass, volume and density. Archimedes' force and floating (lower secondary; the exact year depends on the syllabus in force, so check it) give the full explanation: a pocket of warm air floats in cold air exactly as a cork floats in water.
- **Geography** (*Geografie*) — at **year V**, the Earth's atmosphere: air temperature, pressure, winds and local winds, including the sea breeze. Convective clouds, day and night breezes and the general circulation of the atmosphere are the same physics at a different scale. Mantle currents that move tectonic plates are convection too — there the link is an *application* of the idea, not a physics syllabus entry.
- **Mathematics** (*Matematică*) — calculating a difference and a rate (degrees per metre) at **years V–VI**, alongside percentages; plotting points on a coordinate system at **year VII**; and at **year VIII**, the gradient of a line and the first-degree function give the very language of the vertical thermal gradient.
- **Technology education and practical applications** — why the radiator sits low and under the window, why a water heater warms from the base, why a house's loft is warmer than its cellar. This is a direct application of the measured result, not a separate lesson.
- **Cross-curricular** — the difference between a quantity that grows without limit and one that reaches equilibrium, the care not to contaminate your own measurement, and correcting a phrase almost everyone gets wrong.

The core skill is **linking the shape of a graph to a physical story**. Younger students can usually say "warm air rises"; far fewer can look at a curve that rises and then flattens and say why it flattened. That is the target.

## 2. Time plan

The in-app wizard has eight steps — two of them written predictions — and takes about twenty minutes, as a lesson opener. What turns the activity into a genuine investigation is the vertical profile — temperature measured from the floor up to arm's reach — and that needs a second lesson.

| Route | Time | What happens |
|---|---|---|
| **Short** | **20 min** | Exactly the app's steps: the prediction with drawn arrows (3), the reference at desk height (4), the sensor above the heat source (6), the second prediction and the return to the desk (4), how sure we are, read aloud (1), the conclusion (2). |
| **Full**, lesson 1 | **50 min** | The written prediction with arrows (8), the introduction and the vocabulary — conduction, convection, radiation (4), the reference at desk height, with the boxes side by side (5), above the wet and the dry source, rising to 50 cm and the paper strips (14), the second prediction, the return to the desk and drawing the closed loop (8), how sure we are (4), the conclusion and the written sentence (7). |
| **Full**, lesson 2 | **50 min** | The vertical profile: five heights marked with tape — 0, 50, 100, 150 and 200 cm —, four minutes each (20), the class's shared graph on the board and its gradient (15), the written conclusion (15). |
| Project, lesson 3 | 50 min | Applications and modelling: calculating the density of air at the two measured temperatures, the day and night sea breeze, the hot-air balloon, mantle currents, and a one-page written explanation. |

The full route means **two 50-minute lessons** — an hour and forty minutes — and the project route adds the third, so **three 50-minute lessons**, two and a half hours of contact time. The short route fits comfortably into one lesson alongside other work and needs no preparation beyond filling a container with warm water.

**Give the sensor its time — and fill it**

The BME280 has a small thermal mass, but it is not instant: allow **60–120 seconds** for a temperature reading to settle after the sensor is moved, and considerably longer for the humidity channel, which is the slowest of the three — often three minutes. Plan for this dead time instead of fighting it.

While the sensor settles, students *always* have a task: they draw the predicted arrows for the next position, copy the previous reading into the table, or answer a question written on the board. Thirty students watching a number crawl upwards is a discipline problem waiting to happen.

## 3. Materials

| What you need | Per group | Notes |
|---|---|---|
| MagicBox with BME280 | 1 | Temperature and humidity are used; pressure serves as the control. A long USB cable, or a WiFi connection, is what makes the height series possible. |
| Laptop with the app | 1 | Stays on the desk; only the sensor travels. |
| USB extension cable, 2–3 m | 1 | Strongly recommended. Without it the sensor reaches neither arm's-reach height nor the radiator. |
| **Container of warm tap water, about 45 °C** | 1 | **The default and safest heat source**, not a fallback — and the one the app proposes. It is gentle, burns nothing, never goes out, radiates little (so the sensor measures almost only the rising air), and works in any room. A wide container (a bowl) gives a stronger column than a mug. The teacher fills it from the tap, **never from a kettle**, and sets it on a stable surface, away from the edge of the desk and from the laptop. **It is a wet source**: humidity rises above it. |
| Desk lamp with a filament or halogen bulb | 1 (optional) | A strong alternative, but **hot**: it burns on contact and can scorch paper, and the bulb also warms the sensor directly, by **radiation**, not only through the rising air. A modern LED lamp gives almost no heat — test it beforehand. A dry source: humidity falls above it. |
| Access to a radiator | shared | An excellent source, and more convincing because it is the real thing. Only if the heating is on. |
| Tape measure or metre rule | 1 | For the height of every reading. Heights are *measured*, not guessed. |
| Sticky tape and a marker | 1 set | For fixing the sensor and marking the heights on the wall or the door frame. |
| Step stool or classroom step | shared | For the highest reading only, and only used by an adult or under direct supervision. |
| Thin paper strips or foil | 3–4 | Cut to about 15 cm and taped to a pencil; they make the invisible current above the source visible. Cheap and memorable. |
| Ordinary thermometer | 1 (optional) | A useful cross-check, and a good prompt for a discussion of why two instruments do not read identically. |
| Tray for the container of water | 1 | The water sits on the tray, on one side of the desk; the box and the laptop sit on the other. |

**Preparation the day before:** decide which heat source you will use and *test it*. Many rooms now have LED lighting that produces no usable heat, and discovering that in front of the class is painful. If you are using the radiator, confirm the heating will be running at that time. Cut the

paper strips in advance. Mark the wall or a door frame with tape at the heights you will measure — 0, 50, 100, 150 and 200 cm — so that every group uses the same heights and the class data can be pooled. Finally, **close the windows at least half an hour before the lesson**: a draught destroys the vertical gradient completely, which is a result worth discussing in itself, but not one you want to obtain by accident.

## 4. Scientific background for the teacher

**Why heating makes air rise**

Air is a gas, so its density follows from the ideal gas equation of state. With $\rho$ the density, $p$ the pressure, $M$ the molar mass of air (0.0289 kg/mol), $R$ the gas constant and $T$ the absolute temperature:

$$\rho = \frac{pM}{RT}.$$

Within a room the pressure is essentially uniform, so density is inversely proportional to absolute temperature. Warming air from 20 °C (293 K) to 30 °C (303 K) reduces its density in the ratio 293/303, that is by about 3.3%: from roughly 1.20 to 1.16 kg/m$^3$.
A pocket of this lighter air, surrounded by cooler, denser air, receives from the surrounding air an upward force equal to the weight of the air it displaces, while its own weight is smaller. The net upward force per unit volume is

$$f = (\rho_{\text{cold}} - \rho_{\text{warm}})\, g.$$

For the 3.3% difference above, that is 0.04 kg/m$^3$, this comes to $0.04 \times 9.81 \approx 0.4$ N per cubic metre — very small, but nothing opposes it except internal friction, so the pocket accelerates upwards. Cooler air flows in underneath to take its place, is warmed in turn, and the process sustains itself: a **convection current**.
**The correct phrasing, to be used all lesson:** not “heat rises”, but **heated air rises**, because it has become less dense. Heat is not an object and has no preferred direction — it moves down without any trouble through a metal bar.

### 4.1 Why the temperature above the source levels off

Students expect the temperature to keep climbing. It does not, and the reason is a balance between two rates. The rising column of warm air brings heat to the sensor, while the sensor loses heat to the surrounding air and by radiation. When the two rates become equal, the reading settles — an *equilibrium*, not a stop.

The same reasoning explains why a room with a radiator reaches a steady temperature instead of heating up forever, and it is one of the most transferable ideas in all of school physics. It is worth stressing: a number that stops changing does not mean nothing is happening any more, it means two things are happening at the same rate.

### 4.2 Convection or radiation? How to tell them apart

The sensor receives heat by two routes. Warm air rising past it heats it by **convection**; a hot source also heats it directly, by infrared **radiation**, which needs no air and goes in every direction, not only upwards. A filament bulb radiates strongly: above it, a good part of the reading is radiation, not convection. That is why the default source is the container of warm water: at about 45 °C it radiates little, and what the sensor measures above it is almost only the rising air.

With a lamp, the two separate simply: **hold the sensor at the same distance, but beside the bulb**, level with it. There it receives roughly the same radiation, but not the column of warm air. The difference between the reading above and the reading beside is the convection part. If the

two come out almost equal, your source is more a radiator than a convection engine — and that is a good result to discuss.

### 4.3 What humidity does above the source — it depends on the source

This puzzles students, and it often puzzles teachers too, so it is worth stating precisely. The answer depends on what kind of source you use.

**Above the container of warm water, humidity rises**, usually by 10–30 percentage points. The warm water evaporates, and the air rising off it carries vapour with it: the *numerator* of the ratio changes, and faster than the denominator. This is exactly the *Water Cycle* activity. The trap is the conclusion "warm air is more humid": it was not the heat that brought the humidity, but the water.

**Above a dry source — radiator, lamp, the laptop's air vent — humidity falls**, and this is where the idea of relative humidity shows. The sensor reports **relative** humidity: the amount of water vapour present, as a percentage of the maximum the air could hold at that temperature. The maximum is given by the saturation vapour pressure $e_s$, which rises very steeply with temperature. A convenient approximation is the Magnus formula, with $T$ in degrees Celsius and $e_s$ in hPa (the app uses the same constants):

$$e_s(T) = 6.1094 \exp\left(\frac{17.625\,T}{243.04 + T}\right).$$

At 20 °C this gives about 23.3 hPa, and at 30 °C about 42.3 hPa — 1.8 times as much, an increase of over 80% for ten degrees. The amount of water in the air above the dry source has not changed at all, but the denominator of the ratio has nearly doubled, so the relative humidity falls to about 55% of its earlier value — almost a halving.

The classroom phrasing is: *the air did not become drier, it became thirstier.* This is exactly the mechanism behind fog burning off in the morning sun, and behind the dry feeling of heated buildings in winter.

### 4.4 Where else this happens

Convection is not a laboratory curiosity. The same buoyancy drives the thermals that glider pilots and birds of prey use, the cumulus clouds that form when a rising pocket of air cools to its dew point, the day and night sea breezes on the coast (onshore by day, when the land warms faster; offshore by night, when it cools faster), the great Hadley circulation of the tropics, and — on a timescale of hundreds of millions of years, and with rock instead of air — the mantle currents that move continents. Made explicit, this ladder, from a desk lamp to plate tectonics, is what lifts the activity above a simple demonstration.

## 5. Lesson flow

The in-app wizard has **eight steps**: (1) the introduction — what convection is, and the heat source; (2) a **written prediction** — the arrows of the loop, and what the humidity does above the warm water and above a dry source; (3) the first reading, at desk height; (4) the experiment above the heat source; (5) the second prediction — will the temperature fall faster or slower than it rose, and why does it fall at all; (6) the return to the desk, with the loop drawing; (7) **How sure are we?** — boxes side by side and a repeated reading; (8) the conclusion, which quotes the group's prediction beside its numbers. Below is what the teacher does around them. The times are for the full 50-minute lesson; the 20-minute short route follows the same steps, at the times in the time-plan table.

1. **The written prediction, with drawn arrows, before any measurement (8 min).** Hand out the sheets and ask for **three** written commitments, each with a reason.

First, the **arrows**: on the sketch of the room on the sheet, every student draws how they think the air moves around the heat source. Ask explicitly for a **closed loop** — if air rises somewhere, it must come down somewhere else. Arrows drawn *before* measuring are the best evidence of what the class believed at the start, and comparing them with the drawing made at the end is the learning itself.

Then, two questions: *is the air near the ceiling warmer, cooler or the same as the air near the floor, and by how many degrees?* and *when we hold the sensor above the container of warm water, and then above the radiator (or the lamp), what does the humidity do in each case — rise, fall or stay the same?* The second question splits almost every class, and the fact that the two sources give opposite answers is the engine of the lesson. Do not resolve it. Write the tally of votes on the board and leave it there until the end. Groups type the same predictions into the app, at the prediction step.

2. **Introduction on screen and vocabulary (4 min).** Use the wizard's first step to fix the three words: *conduction* through a solid, *convection* through a moving fluid, *radiation* across a gap. Ask for one example of each before moving on. The activity is opened from the **Activities** page, with the **Start** button on the **Air Convection** card.

**Technical note — the sensor and the buttons, before the first reading**

**The sensor.** BME280, which gives temperature, relative humidity and pressure from a single chip. Nothing extra is plugged in. The cards on screen are **Temperature** and **Humidity**, with their chart underneath, its time axis starting at 0:00 on every step.
**How long to wait. 60–120 seconds** for temperature after every move; humidity is slower, often three minutes. A reading taken ten seconds after moving is the reading of the *previous* place. Give students a task for the duration of the wait — see the teaching tip in the time plan.
**The buttons you will use.** The activity moves forward with the green button at the bottom right, and **its label changes at every step**, because it echoes what you have just finished: **Got it, let's go!**, **My prediction is written**, **I noted the starting values**, **I watched the changes**, again **My prediction is written**, **I sketched the air circulation**, **Understood**, and at the last step **Finish**. The **Back** button, on the left, goes back one step. Above the chart sit **Smooth: on/off**, **Pause** — which freezes the chart, useful when you want to show the class the shape of the curve — and **Clear**.
**Capture.** On the three measurement steps, beside the cards, sits the **Capture** button: it keeps the mean of the last five seconds, and the conclusion quotes the values captured this way. Until it is pressed, the green button stays pale and a tip appears; it blocks nothing. The chart, the journal and the export buttons appear only on the measurement steps (the journal also at the conclusion); the prediction steps show only the answer box. What students type in the journal, predictions included, is saved to disk (the `logs` folder); at **Finish** the activity's Excel file, with its **Journal** sheet, is written there too.
**Exporting the data.** Two buttons appear on the measurement steps: **PNG** saves the charts as an image — exactly what you need for the shape of the curve — and **Excel** saves that step's raw, unfiltered readings to an `.xlsx` file, with an `elapsed_s` column. **There is no CSV button inside the activity.** The whole recording is taken from the **Data & Export** page of the application, with the **Excel** button on the box's row (it opens the **Export data** window, where CSV can also be chosen). Save it using the pattern `class_date_activity` — for example `7A_2026-09-14_convectie_aer`. Note on the sheet the heat source used, the height of each reading, whether the heating was on and whether the windows were closed.

3. **The reference at desk height (5 min).** The sensor sits on the desk, well away from lamps, radiators, windows and students. Wait the 60–120 seconds the app asks for, then students press **Capture** and record the temperature, the humidity and a note on any air currents they can feel. If two groups can, they put their boxes side by side now: the difference between

them is used at the **How sure are we?** step. Tell them plainly: **a measurement without a reference is not a measurement.**

4. **Above the heat source (14 min).** The sensor is held or taped **10–15 cm above** the container of warm water — the default source —, and then, if you have one, above a dry source: the radiator, the lamp or the laptop's air vent. Students watch the *graph*, not the number, because the shape is what matters. Ask them aloud, while the curve climbs: *is it rising in a straight line, or is it bending?* When it flattens, ask why it stopped — the source is still there, still warm.

   Then have a student slowly raise the sensor to 30, 50 and 80 cm above the source and describe what the reading does. Finally, hold the paper strips just above the source: the current becomes visible, and the link between the invisible flow and the number on the screen is made on the spot.

   If you use the lamp, also take the **side reading**: the sensor at the same distance from the bulb, but to one side, level with it. It receives the same radiation but not the column of warm air; the difference from the reading above is the convection part.

   Students press **Capture** once the curve has levelled off above the source. If they measure above two sources, they capture at each: the app's conclusion quotes the last value captured, so the first one goes on the sheet.

**Technical note — right at the step above the heat source**

**The humidity card moves in different directions depending on the source, and it is not a fault.** Above the container of warm water it **rises**, by 10–30 percentage points: the air rising off the water carries vapour with it. Above a dry source it **falls**, by 10–25 points: the air has not lost water, it has gained the capacity to hold more, so the displayed percentage falls. **Humidity is the sensor's slowest channel** — allow three minutes for it to settle, not one. Above the water, hold the sensor beside the visible steam, not in it: droplets block it at 100%.

**Pressure is your control.** The BME280 also gives pressure, and it *does not* respond to the heat source: it stays at the room's value, somewhere between 950 and 1030 hPa, while the temperature climbs by whole degrees. Show the class this. It proves that the sensor is not simply reacting to being held or moved, but to the warm air itself — the cheapest control in the whole activity. The activity has no pressure card: read it from the **Live Data** page.

**The column of warm air is narrow.** Above a lamp, a sideways shift of a few centimetres changes the reading a lot. This is not noise: it is the real shape of the current. Fix the sensor with tape if you want a comparable series.

**The sensor never touches the source.** Contact turns the experiment into conduction, gives an alarming reading and can damage the sensor. Keep the 10–15 cm gap.

5. **The second prediction and the return to desk height (8 min).** Before the move, the app asks for the second written prediction: will the temperature fall faster or slower than it rose, and why does it fall at all? Bringing the sensor back then lets students watch the fall and compare it with the rise. Ask which was faster, and why. **Watch for a common misreading:** the temperature falls not because "the cold air sank onto the sensor", but because *the sensor* has left the warm column and is cooling back to the room's air temperature. The air did not move to the sensor; the sensor moved. Meanwhile, everyone **redraws the loop** on the sheet, now with what they saw: arrows rising above the source, spreading out under the ceiling, descending on the far side of the room, and returning near the floor. **Insist on the closed loop** — a single arrow pointing up is the most common incomplete answer. Compare this drawing with the one made at the start of the lesson. When the sensor is back on the desk and the number holds still, students press **Capture**.

6. **How sure are we? (4 min).** This step of the app has no chart: it asks for two checks. *Side by side:* two boxes set next to each other on the desk for two minutes do not show exactly the same number; the difference between them, often a few tenths of a degree, is how far a comparison between boxes can be trusted. *Twice:* two readings at the same place, thirty seconds apart, after the number has stopped rising. If the spread at one place is bigger than the difference between places, the honest conclusion is "we could not detect a difference with our setup", not "there is no difference". And a window left ajar can turn the whole result upside down: look for the draught before blaming the box.

7. **The conclusion and the compulsory sentence (7 min).** Return to the tally of votes on the board and settle the humidity question with the "thirstier, not drier" explanation. Then ask every student for the **closing sentence on the sheet**, in their own words, with a single formal requirement: the sentence's subject must be **heated air**, not heat. "Heated air rises because it is less dense" passes; "heat rises" does not, and gets rewritten on the spot. Close with the app's conclusion step, which quotes their prediction beside their numbers — temperature and humidity at the desk and above the source, the temperature difference, the temperature on return, and the density of the air in the two places, computed with the ideal gas law at standard pressure — and whose green button reads **Finish**, and its outward links: clouds, the sea breeze, the radiator under the window, the mantle that moves continents.

8. **The vertical profile (lesson 2, 20 min of 50).** Using the heights marked with tape — 0, 50, 100, 150 and 200 cm — the sensor is held at each for at least 90 seconds and the temperature is recorded; with the move and the recording, allow four minutes per height, so $5 \times 4 = 20$ minutes. The result, plotted as temperature against height, usually gives a difference of one to three degrees between the floor and the highest point, considerably more if the heating is on. Pool the data from every group. This is the most convincing evidence in the whole activity, precisely because it is measured in their own ordinary classroom, where nothing special is happening.

**The investigation, in one table**

**The question.** How does the air temperature change with height, and by how much?
**The hypothesis.** Written on the sheet **before** any measurement, in two forms, both given: **drawn arrows** on the sketch of the room, with a closed loop, and a **number with a unit** for the difference between the floor and the highest point. Neither is changed afterwards.
**What I change.** A single thing: the **height** of the sensor, measured with the tape measure from the floor.
**What I measure.** Temperature, in °C, to one decimal place, at every height. Humidity is recorded at the positions above the two heat sources, where it is precisely the question; it is not needed in the vertical profile.
**What I keep the same.** The same box from start to finish, the same corner of the room, the windows and door closed, the same waiting time at every height, the same person reading the screen, and nobody leaning over the sensor.
**How many repeats.** Two readings at the **floor** and two at the greatest height, thirty seconds apart, *after* the number has stopped climbing — you do not wait another two minutes for each one. At the heights in between, a single reading after settling is enough: those give the shape of the profile, not the spread.
**What makes a result valid — and what makes it an artefact.** Compare the spread (maximum minus minimum) of the two floor readings with the difference between the floor and the highest point. **If the spread at a single level is larger than the difference between levels, the correct conclusion is that the difference is not detectable with your setup — not that it does not exist.** These are two different statements, and the distinction between them is exactly what is learned here. The remedy is not to hunt for better numbers in the table, but to increase the signal (switch the heating on, measure

higher) or reduce the noise (wait longer, close the windows). And if the profile comes out *reversed* — colder higher up — look for a draught before blaming the group: a window left ajar can flip the result completely.

# 6. Analysis after the activity

## 6.1 What the data should look like

| Situation | Typical reading | Interpretation |
|---|---|---|
| Reference, desk height | 20–23 °C, 40–55% RH | The value everything else is compared against. |
| 10–15 cm above a container of water at about 45 °C | +1 to +5 °C above the reference (more with a wide container) | Gentle, broad and very stable; almost pure convection. The best-suited source for a class. |
| 10–15 cm above a filament lamp | +4 to +12 °C above the reference | The column is narrow and hot; a small sideways move changes the reading a lot. Part of the reading is radiation straight from the bulb. |
| The same distance, beside the lamp | less than above | Radiation without the warm column; the difference from the row above is the convection part. |
| 10–15 cm above a radiator | +3 to +8 °C above the reference | Broader and steadier than a lamp's column. |
| At 50 cm above the source | +1 to +3 °C above the reference | The column has mixed with the room's air and cooled. |
| Humidity above the container of warm water | higher by 10–30 percentage points | The rising air carries vapour off the water: water has been added. |
| Humidity above a dry source (radiator, lamp) | lower by 10–25 percentage points | Relative humidity, not water content: the air has become thirstier. |
| Floor to 200 cm, heating off | 0.5–1.5 °C warmer at the top | Small, but consistently in the same direction. |
| Floor to 200 cm, heating on | 2–5 °C warmer at the top | The gradient students can then feel with their hand. |
| Settling time after a move | 60–120 s for temperature | Humidity is slower, often three minutes. |
| Pressure | 950–1030 hPa, unchanged | The control: it does not respond to the heat source, so the sensor is not merely reacting to being moved. |

Absolute values depend heavily on the room, the source and the season. Compare *directions* and *shapes* between groups, rather than hunting for a correct number. A group whose lamp gave only two degrees has not failed: they have an LED lamp, and that is a result.

## 6.2 From profile to gradient, step by step

1. Enter, for each height, the value in metres and the temperature in °C, to one decimal place.
2. Plot temperature (vertical) against height (horizontal), scaling the temperature axis *only* over the range of the data — an axis starting at zero flattens the whole effect. The axes are prepared on the sheet for exactly this reason.
3. Calculate the gradient: the top temperature minus the floor temperature, divided by the height difference. Worked example, with typical figures: $22.4 - 20.4 = 2.0\,°\text{C}$ over a height of 2.00 m, so $2.0/2.00 = \mathbf{1.0\,°C\ per\ metre}$.
4. Use the gradient to predict: in a sports hall 4 m high, $1.0 \times 4 = 4.0\,°\text{C}$ between floor and ceiling. Then ask for *one reason* the prediction might be wrong. The honest answer is that the gradient is not constant over the whole height, and a sports hall is not a sealed box.
5. **Extension:** calculate the density of air at the two measured temperatures, with $\rho = pM/RT$ and the pressure given directly by the BME280, express the difference as a percentage, and discuss how such a small difference can drive a visible current.

## 6.3 Questions to ask afterwards

1. The temperature above the source rose quickly and then flattened. Why did it stop rising, if the source was still on?
2. Above the radiator the humidity fell, but nobody removed any water from the room. Explain, using the idea of relative humidity, how both statements can be true at the same time.
3. Above the container of warm water the humidity rose, and above the radiator it fell, even though both warm the air. What does the container have in addition? Can you still say, after this, that "warm air is more humid"?
4. When you brought the sensor back to the desk, the temperature fell. A classmate says: "the cold air sank onto it". What actually moved?
5. Radiators are almost always fitted low on the wall and under a window. Use your profile graph to explain why that is the sensible spot, and what would happen if they were mounted near the ceiling.
6. You drew arrows showing air rising above the source. Where does that air go afterwards, and how do you know it must come down somewhere?
7. A hot-air balloon rises even though it contains the same air as the sky around it. Using density, explain what the heating actually changed.
8. Someone says "heat rises". Correct the sentence and explain in one line why the correction matters.

## 6.4 Ideas for assessment along the way

- **Low stakes:** the student completes a printed cross-section of the room with arrows for the convection loop and labels three points "warm air rising", "cooling at the top" and "cool air sinking". No number is required.
- **Standard:** a short write-up covering the reference reading, the reading above the source, the temperature difference, a sketch of the graph's shape, and a paragraph explaining both why the temperature flattened and why the humidity rose above the water and fell above the dry source. The closing sentence's subject must be **heated air**.

- **Extension:** the student plots the class's shared profile, estimates the gradient in degrees per metre, uses it to predict the difference in a four-metre sports hall, and names one reason the prediction might be wrong. In addition, they calculate the density of air at the two temperatures with $\rho = pM/RT$, give the difference as a percentage, and comment on the ratio between the tiny cause and the visible effect.

### Common pitfalls and misconceptions

**"Heat rises."** Heat does not rise; *warm air* rises, and it does so because it has become less dense, not because heat has any preferred direction. Heat moves down through a metal bar without any trouble. This phrasing correction is worth making explicit and repeating, and the closing sentence on the sheet is where it is checked: the subject must be the air, not the heat.

**"The air dried out above the lamp."** The most common misreading in the activity. The water content has not changed at all; only the percentage of a new, larger maximum has fallen. Use the phrase "thirstier, not drier".

**"Warm air is more humid."** The opposite misreading, which appears above the container of warm water: the humidity rises, and students put it down to the heat. It was not the heat that brought it, but the evaporating water. The same warming without water (the radiator) lowers the percentage.

**"The cold air sank onto the sensor."** On the return to the desk the temperature falls because the sensor has left the warm column and is cooling back, not because cold air moved towards it.

**Radiation taken for convection.** Above a filament bulb, part of the heating comes straight from the bulb, by radiation. The side reading, at the same distance, shows how much.

**An LED lamp that produces no heat.** Modern classroom lighting and desk lamps can be nearly cold. Always test the source beforehand; the container of water at about 45 °C is the recommended source and the one the app proposes.

**Touching the source with the sensor.** Contact turns the experiment into conduction, gives an alarming reading and can damage the sensor. Enforce the 10–15 cm gap.

**Draughts.** An open window, an air-conditioning unit or a door left ajar destroys the vertical gradient completely, and can even reverse it. Close everything, and if a group gets an inverted profile, look for the draught rather than blaming the group.

**Breathing on the sensor.** Exhaled air is at 34 °C and nearly saturated. A student leaning in to read the number more closely produces a dramatic and entirely false result. Set the rule from the start — "nobody leans over the sensor" — and if it happens anyway, use the moment: it is a perfect illustration of contaminating a measurement.

**Reading before settling.** A reading taken ten seconds after the sensor is moved is the reading of the previous place. Enforce the timer.

**Expecting a large number.** A one-degree difference between floor and ceiling feels disappointingly small to a thirteen-year-old. Reframe it: this room is not designed to have convection currents, and the instrument found one anyway.

### Safety and classroom management

**The default heat source is warm water, not a hot lamp.** You fill the container yourself with warm tap water, about 45 °C — **never water from a kettle** —, set it on a tray, on a stable surface, away from the edge of the desk and away from the laptop and sockets. Water and electricity do not share a work surface. **The box is not waterproof**: it is held above the water, never in the steam and never where it can be splashed. Wipe up any spill immediately. Students do not carry the container around the room.

**Hot lamps.** Filament and halogen bulbs reach temperatures that burn on contact and can

scorch paper. Keep the paper strips, the sheets and the cables clear of the bulb, never cover a lamp, and let it cool before packing it away. Students hold the sensor above the lamp by its cable or its housing, never by reaching across the bulb.

**Radiators.** School radiators can be hotter than expected. Check the surface yourself first, and tell students to hold the sensor above the radiator without leaning on it.

**Working at height.** The highest reading is the activity's only real hazard. Use a proper step, on level ground, never a chair or a desk. In most schools, an adult takes the highest reading. If you do not have a safe step, simply stop the profile at the highest height a standing student can reach with an arm raised — the trend is already clear by then.

**Cables.** A 3 m USB cable crossing the room is a trip hazard. Tape it down, route it along a wall, and brief the class before anyone stands up.

**Managing the class.** Students moving around the room with a sensor on a long cable need a set route and a single named carrier per group. Agree before the lesson that the laptop stays where it is.

### Adaptations for pupils with special educational needs

**Only up/down and warmer/cooler.** This is the support route, and it reaches the same physical conclusion. The student calculates no difference and records no decimal: for each height they tick an arrow up or down and circle "warmer" or "cooler" relative to the desk. The sheet comes with the reference row marked and all the heights already printed, for exactly this reason. For extra support, draw two of the four arrows yourself in the prediction box *before* copying the sheet: the student continues them rather than inventing them.

**The density calculation stays, for those who can do it.** Do not drop it from the lesson to simplify the class. Students on the extension route calculate $\rho = pM/RT$ at the two temperatures, express the difference as a percentage, and explain why 3% is enough to set the air moving. The two routes reach the same closing sentence, by different paths.

**Three real roles for students with motor difficulties.** The vertical profile does not require everyone to climb the step. Roles rotate at each height:

- **time keeper** — starts the timer, announces when the 90 seconds are up, and has the right to reject any reading taken too soon; this is the role with the greatest scientific weight in the group, because it decides which reading is valid;
- **screen reader** — says aloud when the number has stopped climbing and dictates the two values;
- **data keeper** — holds the group's table, checks on the spot whether the two repeats at the same level are close together, and says if a reading needs repeating.

The sensor can be raised by a classmate, on the student's instruction: "higher by one hand's width" is an experimental decision too, and whoever makes it is running the experiment.

**Spoken answers and drawing instead of writing.** The hypothesis, the explanation of the flattening and the closing sentence can be given **aloud**, with the teacher or a classmate writing them on the sheet. For students who find writing hard, **the arrow diagram is a fully accepted answer** in place of the paragraph: a closed loop, correctly drawn, shows exactly the same understanding.

**Language and cognitive support.** Write the lesson's four words on the board at the start — conduction, convection, radiation, density — and use them unchanged all lesson. Repeat the correct phrasing every time: *heated air rises*. For students sensitive to crowding, the readings above the source can be taken one at a time, quietly, while the rest of the class works on the loop drawing.

**If things do not go to plan**

**You have no lamp, or no heating on.** Convection works anyway with the container of water at about 45 °C, the recommended source: gentle, safe, steady and available in any room. For the humidity question, though, you also need a *dry* source: the laptop's air vent, while the fan is running, is enough. If you have neither, state the result (humidity falls above a dry source) and refer to the warm-and-dry witness in *The Water Cycle*.

**You have one box and thirty students.** Do the measurement once, as a demonstration, with the screen projected or the class in a semicircle. At each height a different student reads the screen aloud and another confirms the number has stopped; everyone records in their own table. You get a single data set, but the whole class analyses it, and the roles rotate.

**You have twenty minutes, not fifty.** This is exactly the short route: follow the wizard step by step. Keep the drawn arrows, the reference on the desk, the reading above the source, and the closing sentence said aloud. Drop the vertical profile, the paper strips and the density calculation by hand — the app shows it in the conclusion anyway.

**You have no projector.** Semicircle around the laptop, close enough to see the graph, and a named **screen reader** at each position, who says aloud what the curve is doing: "rising fast", "rising more slowly", "it has stopped". The shape of the curve carries very well in words, and describing it aloud is exactly the skill being assessed.

**The sensor will not settle.** This is normal in the first minute and a half. If the number is still drifting after three minutes, the cause is almost certainly a draught: close the windows and the door and wait another two minutes, with the sensor set down, not held in the hand. If it still will not settle, move the setup away from the radiator and the window and start again.

**The box will not connect.** Do not stop the lesson. Use the "What the data should look like" table from this guide as a printed data set: students draw the arrows first, then receive the values row by row, calculate the gradient and make the prediction for the sports hall. The discussion of "heated air rises, not heat" carries on unchanged. Repeat the measurement on another day, after checking the cable, the port selected in the app and, if you are on WiFi, the network.

## 7. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are root domains, not deep links, so they will not rot over time.

- **National Meteorological Administration (ANM)** — `https://www.meteoromania.ro` — search for clouds and breezes on the Romanian coast; the same convection at the scale of a day and a county, with photographs that make the link vivid.
- **PhET Interactive Simulations** — `https://phet.colorado.edu` — search for the simulations on gas properties and states of matter; they let students see the particle motion behind the density change, which they cannot observe directly.
- **NOAA** — `https://www.noaa.gov` — search for atmospheric circulation and thermals, for the step from a classroom column of warm air to global weather patterns.
- **Institute of Physics** — `https://www.iop.org` — search for practical resources on heat transfer; a good source of further low-cost convection demonstrations to run alongside this one.
- **STEM Learning** — `https://www.stem.org.uk` — search for heat transfer and convection for ages 11–14; classroom-tested sheets and a large bank of related activities.
- **United States Geological Survey** — `https://www.usgs.gov` — search for mantle convection and plate tectonics, for the geological end of the ladder that begins with a desk lamp.

## 8. Assessment

**The exit ticket**, identical to the one on the student sheet: *one thing that surprised you today, and one thing you would do differently if you repeated the measurement.* It is collected in the last two minutes of the lesson and says more about the class's understanding than any mark.

For a more formal judgement, use the **shared three-strand rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. At Explorer level, the three strands look like this:

- **Measurement practice** — the student waits for the reading to settle before recording, takes a reference before anything else, measures the heights instead of guessing them, and does not lean over the sensor.
- **Data interpretation** — the student describes the shape of the curve, not just the final value, calculates the gradient with its unit, and distinguishes "the difference is not detectable" from "the difference does not exist".
- **Scientific reasoning** — the student explains the flattening as a balance between two rates, explains the drop in humidity above the dry source without claiming water was lost and its rise above the water through evaporation, and writes the closing sentence with **heated air** as the subject, not heat.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

## 9. The student sheet

The sheet the students fill in is a separate document: `docs/en/student_sheets/air_convection.pdf`. It has two pages — the prediction, the two arrow boxes and the measurement above the source on the first; the vertical profile, the graph, the gradient and the conclusion on the second — so it prints double-sided, one sheet of paper per student. It is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write — and asks for the prediction **before** any measurement.

Three things worth knowing before printing it. **The arrow box comes before any table:** the student draws the predicted loop first, redraws it at the end in the second box, and the two drawings side by side are the evidence of learning. **The tables have their rows already labelled**, with the reference row and all five heights printed, and the graph's axes are already drawn — the student does not lose the lesson drawing axes. The measurement table has one row for the warm water and one for the dry source, because humidity moves in opposite directions above them. Raising the sensor to 30, 50 and 80 cm above the source is described aloud, at step 4, and has no row in the table. Under the table, a short line for **How sure are we?**: the difference between boxes side by side and the two readings at the same place. **The last sentence has a printed start**, *Heated air rises because…*, precisely so that the sentence's subject is the air, not the heat. The tables are not repeated here precisely so that the two cannot drift apart: there is a single printable version, and it is the sheet.